

Advanced (Habanero)

Q1. Factor $x^2 - 4$

- (A) $(x + 2)(x - 2)$
- (B) $x(x - 4)$
- (C) $x(x + 4)$
- (D) $x^2 - 2x + 4$
- (E) $x^2 + 2x - 4$

Q2. Factor $9 - y^2$

- (A) $(3 + y)(3 - y)$
- (B) $(3 + y)(3 + y)$
- (C) $(9 + y)(9 - y)$
- (D) $(9 - y)(9 + y)$
- (E) $(3 - y)(3 - y)$

Q3. What is the factored form of $x^2 - 64$?

- (A) $x(x - 64)$
- (B) $x(x + 64)$
- (C) $(x + 8)(x - 8)$
- (D) $(x - 8)(x - 8)$
- (E) $(x + 8)(x + 8)$

Q4. Solve $2x^2 - 8 = 0$ using difference of squares

- (A) $x = pm1$
- (B) $x = pm2$
- (C) $x = pm4$
- (D) $x = pmsqrt2$
- (E) $x = pmsqrt4$

Q5. Factor $81 - 16x^2$

- (A) $(9 + 4x)(9 - 4x)$
- (B) $(9 + 2x)(9 - 2x)$
- (C) $(9 - 4x)(9 - 4x)$
- (D) $(3 + 4x)(3 - 4x)$
- (E) $(3 + 2x)(3 - 2x)$

Q6. Factor $49 - 9x^2$

- (A) $(7 + 3x)(7 - 3x)$
- (B) $(7 + 3x)(7 + 3x)$
- (C) $(7 - 3x)(7 - 3x)$
- (D) $(7 - x)(7 - x)$
- (E) $(7 + x)(7 + x)$

Q7. Solve $(x^2 - 9) = 0$

- (A) $x = pm3$
- (B) $x = pm9$
- (C) $x = 3$
- (D) $x = -3$
- (E) $x = 0$

Q8. Factor $x^2 - 25y^2$

- (A) $(x - 5y)(x + 5y)$
- (B) $(x - y)(x + y)$
- (C) $(x + y)(x - y)$
- (D) $(x + 25y)(x - 25y)$
- (E) $(x^2 - y^2)$

Open Ended Questions (FRQ)

Q9. Factor $x^4 - 81$ using difference of squares

Q10. Prove that $a^2 - b^2 = (a - b)(a + b)$

Chef's Tips

- Emphasize that factoring is a way of expressing an algebraic expression as a product of simpler expressions
- Encourage students to look for common patterns when factoring
- Remind students that the difference of squares formula only applies when there is a subtraction of two perfect squares